Introduction to Linux Systems

Inter-Process Communication (IPC)

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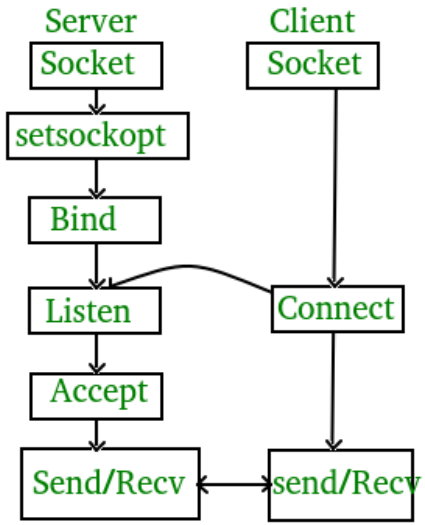
Mini project: Socket Programming in C

- https://www.geeksforgeeks.org/socket-programming-cc/

```
Client:
Hello message sent
Hello from server

Server:
Hello from client
Hello message sent
```

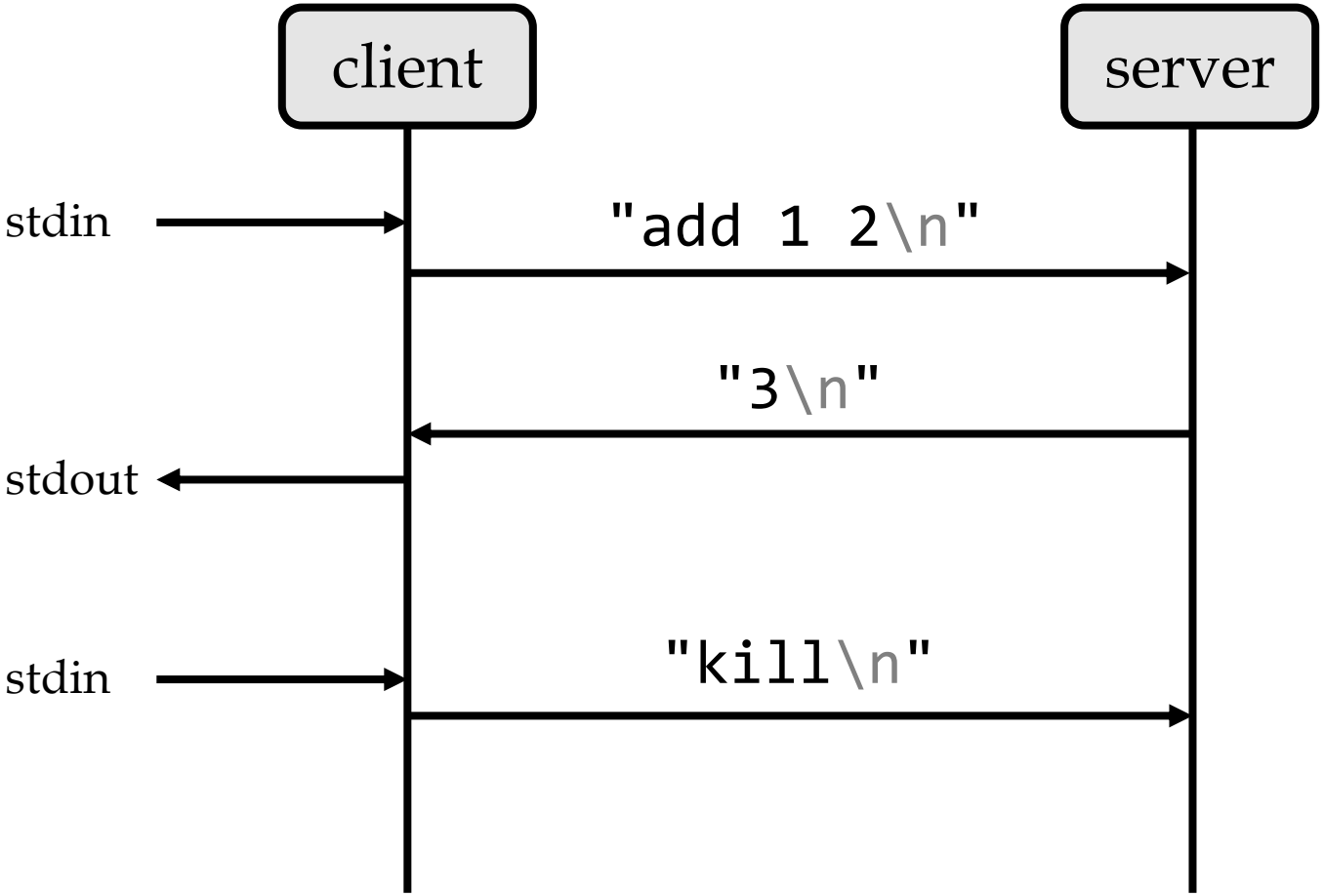


IPC using simple socket

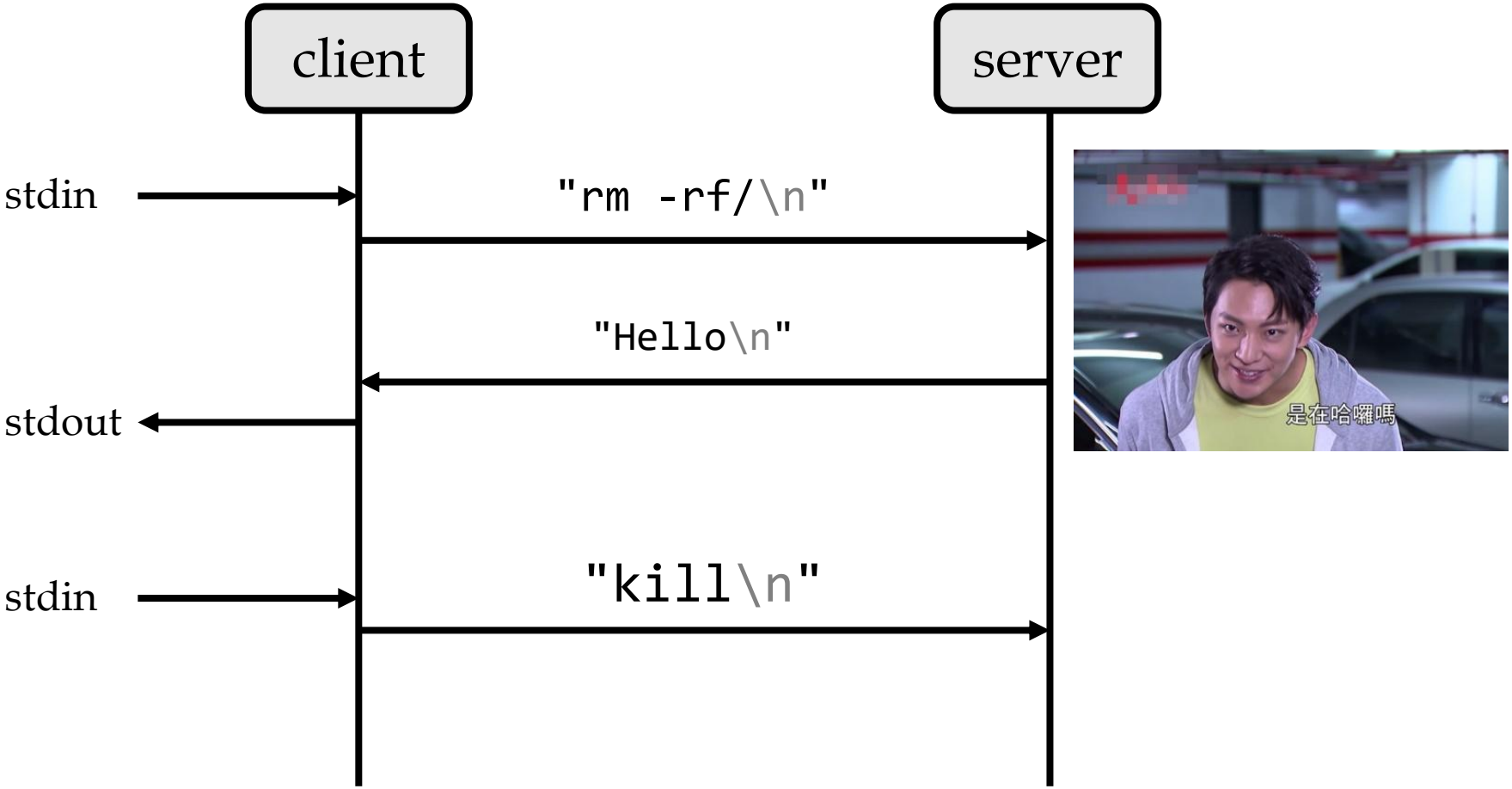
- server
  - supports three commands
    1. "`add a b`": return $a + b$
    2. "`abs a`": retuen $|a|$
    3. "`mul a b`": return $a \times b$
  - When getting unsupported command: return Hello
- client
  - reads the commands until EOF
  - sends the commands to server

- supports command kill

Execution flow (supported)



Execution flow (unsupported)



https://star.setn.com/news/851769

You should pass 4 tasks (60pt)

```
add 1 26
abs -100
mul 33 55
unsupport!!
kill

```

```
$ cat out1
27
100
1815
Hello
```

Input 檔案最後一行是空的

拿回來的答案包含換行

Test by yourself

This also how we test your program

- ans file is provided in ans.zip

$ make

$ ./server & ./client < in > out

$ diff -s out ans

if your output is correct, you will get

"Files out and ans are identical"

Hint: you can add some delay in client to avoid random connection error

Sample of Makefile:

```
all: result1 result2
.PHONY = clean result1 result2
result1: file1

	WHAT_YOU_WANT_TO_DO
result2: file2

	WHAT_YOU_WANT_TO_DO
clean:

	HOW_TO_CLEAN(hint:rm)
```

Operation process and comments (40pt)

- Clear, Simple, and Useful
- Comments in source code

This will be scored by demo video
(video should not exceed 3 min)

Submission

- Upload **F12345678.zip** to moodle
- **F12345678** is your Student ID
- <u>You will not get any scores if you don't follow the submission rules</u>

```
F12345678.zip
└── F12345678/
    ├── in
    ├── ans
    ├── client.c
    ├── server.c
    ├── Makefile
    └── Demo.mp4
```